

Technical Datasheet Input / Output Modules with Modbus TCP Protocol with Ethernet Interface

The IO module communicates via Ethernet Interface. The module is designed to allow interface with other Modbus TCP/IP compatible devices. The module acts as an input/output device between the Modbus TCP/IP Network and the Gateway.

Digital IO module is sturdy, low power usage and is easy to use.

4 Port AI Module with Modbus TCP Ethernet Port: -



The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators.

The design of the modules incorporates '**resettable Fuses**' to safeguard against sudden High Voltage spikes. The board also has '**reverse polarity**' protection for input **Power** connector.

The communication port also has signal isolation and additional Jumper for 'end of line' 120 ohms termination resistance which may or may not be used.

Specifications

General: –

I/O Connectors	2 Pin 5.08 mm pitch pluggable screw terminals.
Dimensions	70 mm L x 110 mm B x 50 mm H
Power	Input Power – 12 – 24 VDC or 24 V AC/DC Typical – 12V DC @ 1Amp
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing

Certifications



AI Inputs: –

Channels	4
Input Signal	4 – 20 mA / 0 – 5 V / 0 – 10 V (jumper selectable)
Accuracy	± 2 % of Full scale
Input Resolution	10-bit /12-bit resolution (optional)
Isolation	Optically Isolated
External Loop Voltage	+ 12 VDC min
AI input impedance	120Ω

Additional Features: -

Input power reverse polarity safety
 ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air EFT
 IEC 61000-4-4, 50A (5/50ms)
 750V isolation.
 CRC Error check.
 Web based Configuration needed for the IO board

Configuration Settings: -

Communication Speed	10/100 Base-T Ethernet compatible Interface
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	(Web Selectable)
Function Code AI	0x03 Read Holding Registers
AI Register Address	10 Bit -1,2,3,4 / 12 Bit - 5,6,7,8.

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	AI 1 – 10 Bit	40001	0X03	RS485	16 Bit Unsigned int
2	AI 2 – 10 Bit	40002	0X03	RS485	16 Bit Unsigned int
3	AI 3 – 10 Bit	40003	0X03	RS485	16 Bit Unsigned int
4	AI 4 – 10 Bit	40004	0X03	RS485	16 Bit Unsigned int
5	AI 5 – 12 Bit	40005	0X03	RS485	16 Bit Unsigned int
6	AI 6 – 12 Bit	40006	0X03	RS485	16 Bit Unsigned int
7	AI 7 – 12 Bit	40007	0X03	RS485	16 Bit Unsigned int
8	AI 8 – 12 Bit	40008	0X03	RS485	16 Bit Unsigned int

Note: -

For MODBUS communications, a **shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

Configuration Guide – Ethernet Modbus TCP/IP IO Module:-

1. Open a web browser (Google Chrome, Microsoft Edge, Firefox, etc.).
2. In the browser's address bar, type the default IP address:
192.168.0.160
3. Press **Enter**. A message will appear saying:
"Please go to <http://192.168.0.160/setup>."
4. **Copy** this link (<http://192.168.0.160/setup>) and **paste** it into a new browser tab.
5. Press **Enter** again to open the **Setup / Configuration Page** of the device.

To Reset the Device to Default Settings

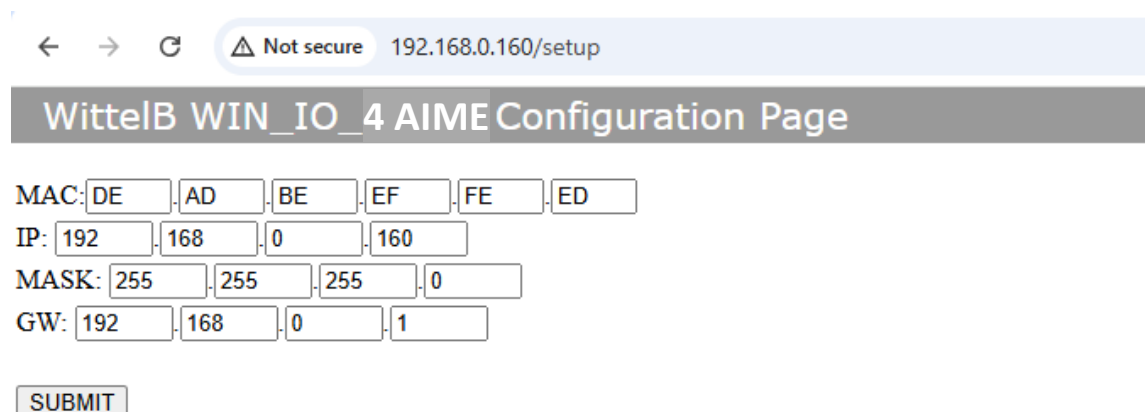
- Place the **jumper** on the designated **Reset position** on the device.
- Press the **Reset button** once.
- The module will reboot and revert to its default IP address: 192.168.0.160.

To Change the IP Address

- On the **Setup / Configuration Page**, enter the **new IP address** you wish to assign.
- Click **Submit** to save the new settings.
- The device will automatically apply the **new IP address**.

Note:

- Ensure your PC's IP address is in the **same network range** (for example, 192.168.0.xxx) when accessing the module.
- After changing the IP, you must use the **new address** to log in again.

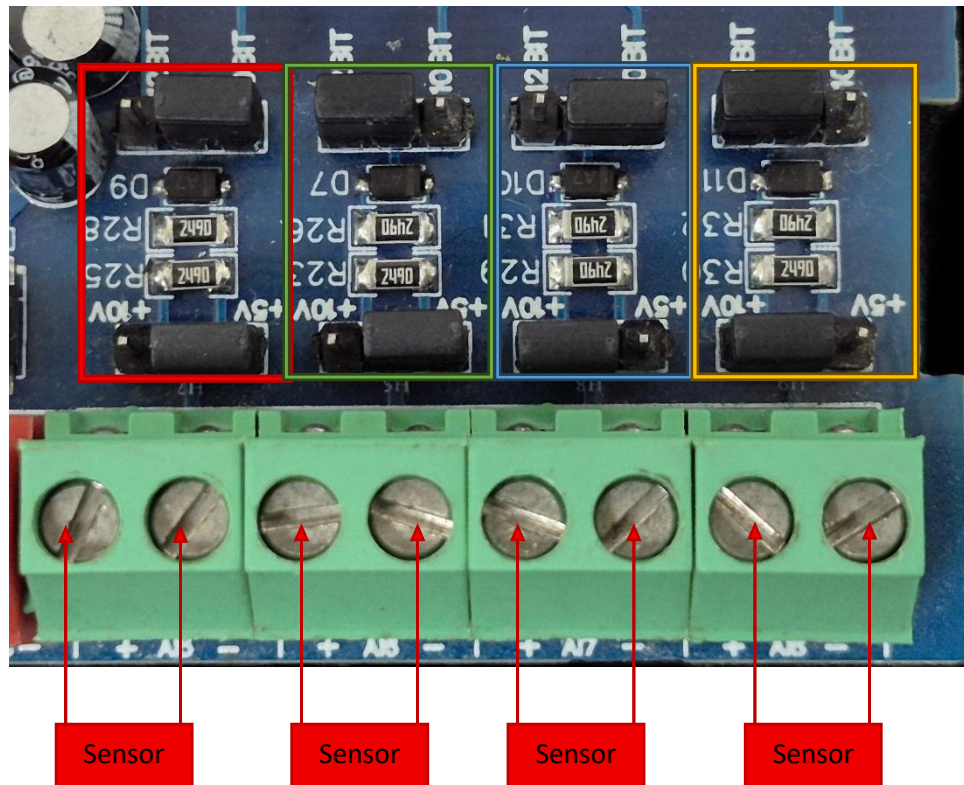


The screenshot shows a web browser window with the address bar displaying "192.168.0.160/setup". The page title is "WittelB WIN_IO_4 AIME Configuration Page". The form contains the following fields:

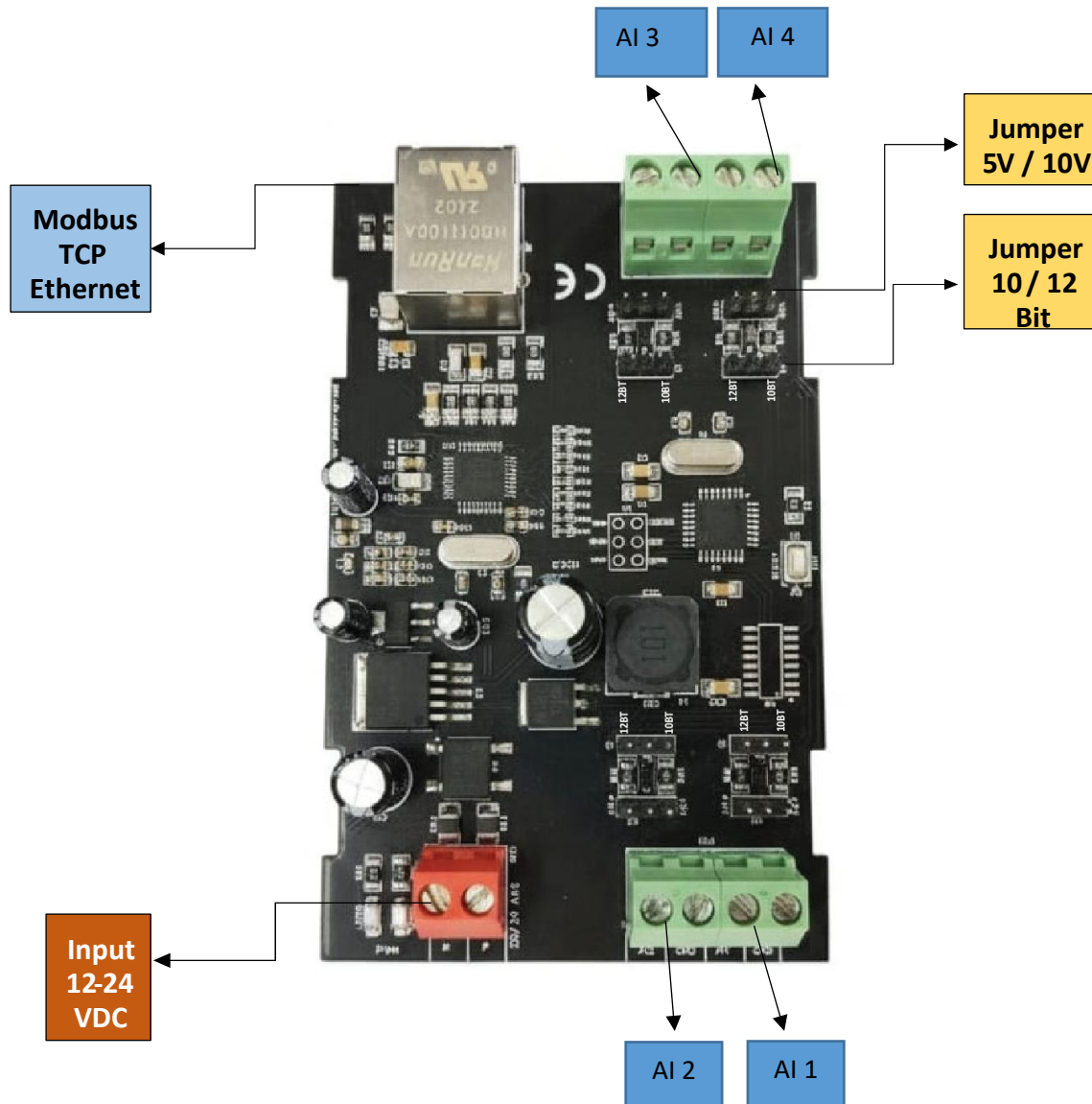
MAC:	DE	AD	BE	EF	FE	ED
IP:	192	168	0	160		
MASK:	255	255	255	0		
GW:	192	168	0	1		

Below the fields is a **SUBMIT** button.

AI



- **Analog Input (AI) Operation:** Each channel provides two terminals: + (Positive) and - (Negative/Common). These are used to read external analog sensor signals.
- **Channel Configuration (Visual Guide):** Each of the four AI channels features two dedicated jumpers to independently set the voltage range and bit resolution. Refer to the color-coded boxes in the board diagram to configure your desired scenario:
 - **Red Box (5V, 10-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **10 BIT**.
 - **Green Box (5V, 12-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **12 BIT**.
 - **Blue Box (10V, 10-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **10 BIT**.
 - **Yellow Box (10V, 12-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **12 BIT**.
- **Sensor Wiring Connections:**
 - The analog sensor's positive signal wire should be connected to the **+** terminal of the desired channel.
 - The sensor's common or ground wire should be connected to the corresponding **-** terminal.
- **Crucial Setup Recommendation:** Always ensure that the voltage jumper setting strictly matches the maximum output range of your connected sensor (either 5V or 10V) to guarantee accurate analog-to-digital conversion and prevent data clipping.



NOTE:-

By default, the Jumper setting on the board is 10 V and 10 Bit.

Contact us: -

Augmatic Technologies Pvt. Ltd.,
Plot no 6, Shah Industrial Estate II,
Kotambi,
Vadodara – 391510.

Email – Sales@wittelb.com

Support: support@wittelb.raiseaticket.com / +91 94038 92805